Worksheet 27 · Perp. & Parallel Lines IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Identify parallel lines from their gradients • Find the gradient of a perpendicular line • Find the equation of a line perpendicular to a given line		
Grade: / 56	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Perpendicular Gradients → Negative reciprocal, $m_2 = \frac{-1}{m_1}$

Alternate Equation of line → $y - y_1 = m(x - x_1)$

Questions *Perp. & Parallel Lines*

Question 1 (3 marks)

The straight line L_1 has equation $x + 2y = 4$
The straight line L_2 passes through the points $(-1, -7)$ and $(7, 9)$

Michael says that the lines L_1 and L_2 are perpendicular.

Is Michael correct?
You must show clearly how you get your answer.

Question 2 (3 marks)

Point A has coordinates $(5, 8)$
Point B has coordinates $(9, -4)$

(a) Work out the gradient of AB.

.....
(2)

The straight line L has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to L.

.....
(1)

Question 3 (4 marks)

The centre O of a circle has coordinates $(4, 7)$

The point A, on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

.....

Question 4 (5 marks)

ABC is an isosceles triangle with $AB = AC$.

B is the point with coordinates $(-1, 5)$
C is the point with coordinates $(2, 10)$
M is the midpoint of BC.

Find an equation of the line through the points A and M.
Give your answer in the form $py + qx = r$ where p, q and r are integers.

.....

Question 5 (5 marks)

ABC is a triangle in which angle $ABC = 90°$

p and q are integers such that

the coordinates of A are $(p, 10)$
the coordinates of B are $(-1, -5)$
the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p =$

$q =$

Question 6 (7 marks)

The straight line **L** passes through point A (– 6, 2) and point B (5, 3)
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B.
The line **M** intersects the line $x = -1$ at point C.

Calculate the area of triangle ABC.

Question 7 (7 marks)

$ABCD$ is a kite, with diagonals AC and BD, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates (−3, 4)

The diagonals of the kite intersect at the point M with coordinates (0, 2)

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D.

(.....,.....)

(.....,.....)

Question 8 (4 marks)

The straight line $\mathbf{L}_1$ has equation $2y = 6x - 5$

The straight line $\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point (9, −1)

Find an equation for $\mathbf{L}_2$
Give your answer in the form $ay + bx = c$

Question 9 (4 marks)

Line **L** has equation $4y - 6x = 33$
Line **M** goes through the point A (5, 6) and the point B $(-4, k)$

L is perpendicular to **M**.

Work out the value of k.

Question 10 (5 marks)

$ABCD$ is a kite, with diagonals AC and BD, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates (−3, 4)

The diagonals of the kite intersect at the point M with coordinates (0, 2)

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D.

Question 11 (5 marks)

$\mathbf{L}_1$ and $\mathbf{L}_2$ are two straight lines.
The origin of the coordinate axes is O.

$\mathbf{L}_1$ has equation $5x + 10y = 8$
$\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point with coordinates (8, 6)

$\mathbf{L}_2$ crosses the x-axis at the point A.
$\mathbf{L}_1$ intersects the straight line with equation $x = -3$ at the point B.

Find the area of triangle AOB.
Show your working clearly.

(Total for Question 24 is 5 marks)

Question 12 (4 marks)

The straight line **L** passes through the points (4, −1) and (6, 4)

The straight line **M** is perpendicular to **L** and intersects the y-axis at the point (0, 8)

Find the coordinates of the point where **M** intersects the x-axis.